Explainable Multimodal Longitudinal Medical Imaging for Disease Progression

NACC V1 — Technical Specification, 5-Member Work Plan, Repository Design & Gantt Roadmap

Project deadline	15 October 2026
Roadmap start	19 September 2026
Team size	5 members / 5 coordinated workstreams
Primary dataset	NACC Freeze 75 — UDS + Mixed Protocol MRI analysis

Goal: one complete, functional and reproducible V1 — not an architecture comparison study.

1. Executive Summary

This specification fixes one implementation path for the V1. The system predicts whether a patient with Mild Cognitive Impairment (MCI) progresses to Alzheimer's disease dementia within 24 months, using longitudinal clinical variables and MRI-derived regional measurements from NACC. The final product is an authenticated Patient Progression Explorer combining prediction, risk trajectory, explainability, dataset audit, and a reproducible research pipeline.

2. Why NACC Instead of ADNI for This Delivery

Criterion	NACC	ADNI	V1 decision
Access timing	Quick Access request; NACC states approved researchers receive data within 48 hours (weekends/holidays excluded).	ADNI notes applications are generally reviewed in about two weeks.	NACC better matches the delivery window.
Longitudinal scale	56,000+ participants and 217,000+ visits; 1–20 visits per participant.	High-quality longitudinal cohort, but smaller and slower access path for this delivery.	NACC provides stronger schedule margin.
Imaging	12,000+ participants with structural MRI; ~2,800 with calculated MRI; mixed-protocol MRI is heterogeneous.	More standardized imaging for many classic ADNI studies.	Use NACC calculated regional MRI measures for V1.
V1 risk	Main risk is matching imaging to clinical visits and missingness.	Main risk is access delay plus full imaging preprocessing workload.	NACC reduces access and 3D-image engineering risk.

Important: this selection is deadline-driven. It does not claim that NACC imaging is intrinsically superior to ADNI. NACC Mixed Protocol MRI is more heterogeneous; therefore V1 intentionally uses quantitative MRI analysis features instead of raw heterogeneous 3D images.

3. Exact NACC Data Request and Governance

- Freeze 75 (September 2026): contains data finalized through 1 September 2026.
- Request `investigator_nacc75.csv` for UDS participant-level clinical/diagnostic data.
- Request `investigator_mri_nacc75.csv` for Mixed Protocol MRI analysis data.
- Participant-level files require Data Use Committee approval and non-commercial research use.
- All collaborators working directly with NACC participant-level data must be covered by the same analysis and DUA conditions.
- No NACC participant-level file, NACCID, processed patient export, or restricted model artifact is committed to Git.

4. Fixed V1 Prediction Task

4.1 Cohort and label

An eligible prediction anchor is a visit at which the participant has MCI, at least one earlier eligible longitudinal observation is available, and sufficient future follow-up exists to determine the 24-month outcome.

- Positive label: MCI at anchor, followed by Alzheimer's disease dementia within 730 days.
- Negative label: MCI at anchor, no Alzheimer dementia during the following 730 days, with sufficient follow-up through the horizon.
- Ambiguous trajectories, insufficient follow-up, or diagnoses that cannot be interpreted reliably are excluded.

- Cognitive status and etiologic diagnosis are both used; the project does not reconstruct diagnosis from cognitive test scores alone.

4.2 Visit construction

- MRI and UDS clinical visits are matched by date because NACC does not automatically pair them.
- Initial matching window: nearest clinical visit within ±90 days of the MRI assessment.
- Each sequence contains a minimum of 2 and maximum of 3 eligible visits.
- Use the three most recent eligible multimodal observations ending at the prediction anchor.
- Retain the real day interval between consecutive visits.

5. Fixed Input Features and Patient Structure

5.1 Clinical input

- Age at visit
- Sex
- Years of education
- CDR global score
- CDR sum of boxes
- MMSE

For each clinical variable, convert NACC missing-value codes to missing values, median-impute from the training set only, and append one binary missingness indicator. Clinical input therefore has 12 dimensions per visit (6 values + 6 masks).

5.2 MRI-derived imaging input

Use quantitative structural MRI analysis variables from the NACC MRI analysis file, such as regional volume and cortical-thickness measurements when available. V1 does not use raw DICOM/NIfTI voxels.

- Remove participant IDs, dates, QC/admin variables, scanner identifiers and non-anatomical metadata from model inputs.
- Retain numerical anatomical MRI features with ≤20% training-set missingness.
- Remove zero-variance features.
- Median-impute from training data only, append feature-level missingness masks, then z-score from training statistics only.
- Preserve a human-readable feature dictionary so explanations map back to anatomical regions.

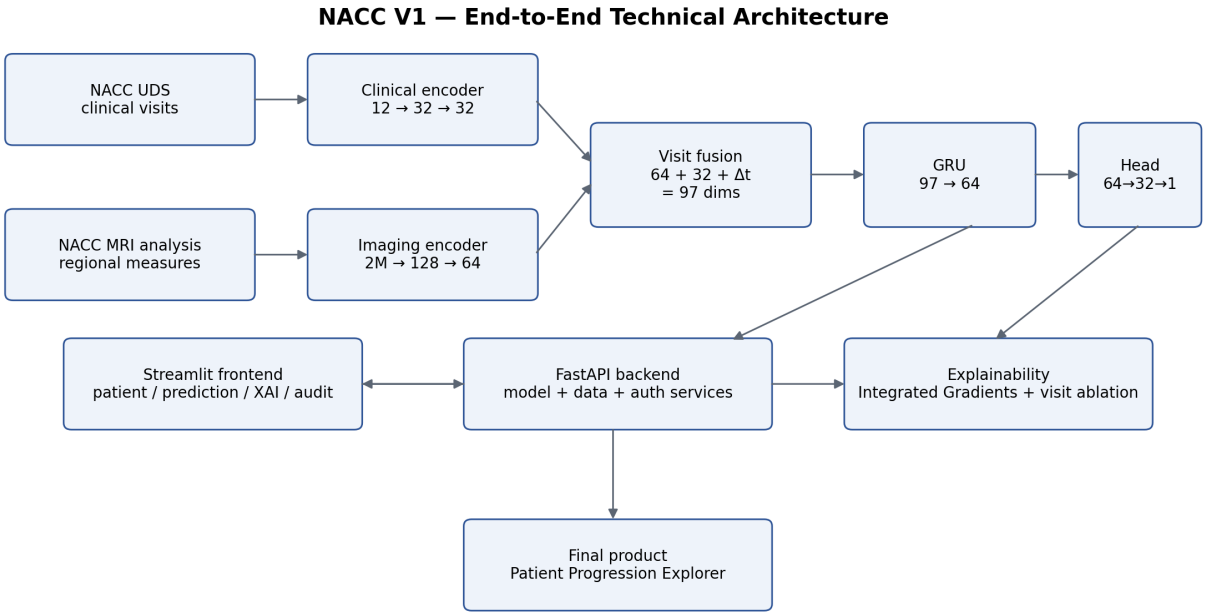
5.3 Final patient tensor contract

```
PatientSequence
  subject_key
  label
  imaging_values        [T, M]
  imaging_missing_mask  [T, M]
  clinical_values       [T, 6]
  clinical_missing_mask [T, 6]
  delta_time            [T, 1]
  sequence_mask         [T]
```

Constraints: $T \leq 3$; M = retained MRI feature count after training-only filtering.

6. Fixed Neural Architecture

Block	Input	Architecture	Output
Imaging encoder	2M	Linear(2M→128) → ReLU → Dropout(0.20) → Linear(128→64) → ReLU	64
Clinical encoder	12	Linear(12→32) → ReLU → Dropout(0.20) → Linear(32→32) → ReLU	32
Visit fusion	64 + 32 + 1	Concatenation of imaging embedding, clinical embedding and Δt	97
Temporal encoder	[T,97]	GRU, hidden=64, one layer, batch_first=True	[T,64]
Prediction head	64	Linear(64→32) → ReLU → Dropout(0.20) → Linear(32→1)	logit



7. Training, Explainability and Evaluation

Item	Fixed V1 choice
Loss	BCEWithLogitsLoss; class imbalance handled with training-set `pos_weight`.
Optimizer	AdamW, learning rate 1e-3, weight decay 1e-4.
Batch / epochs	Batch 32, maximum 100 epochs, early stopping patience 10, random seed 42.
Split	70% train / 15% validation / 15% test, patient-level and stratified by label.
Threshold	Selected on validation data by balanced accuracy; test set is untouched until final evaluation.
Risk trajectory	Prediction head applied at each valid time step; intermediate predictions represent how model evidence changes as visits accumulate.
XAI	Integrated Gradients for feature attribution; leave-one-visit-out ablation for visit importance.
Primary metric	AUROC.
Additional metrics	AUPRC, balanced accuracy, sensitivity, specificity, F1, Brier score, calibration curve.
Reference baseline	One logistic-regression baseline using the final visit only; not an alternative product architecture.

8. Product and Software Architecture

The V1 is a monorepo with separate ML, backend and frontend layers. The backend is the single integration boundary: the frontend never imports ML modules directly.

8.1 Backend

- Framework: FastAPI.
- Services: authentication, patient data access, model inference, explanation generation, dataset audit.
- Authentication: email/password, Argon2 password hashes, JWT tokens, SQLite user database.
- The authentication database stores user accounts only; it never stores NACC participant data.

8.2 Frontend

- Framework: Streamlit.
- Pages: Login, Patient Search, Patient Overview, Prediction, Explainability, Dataset Audit.
- The application displays internal pseudonymous `subject_key` values, never raw NACCIDs.

8.3 API contract

```
POST /api/v1/auth/register
POST /api/v1/auth/login
GET  /api/v1/patients
GET  /api/v1/patients/{subject_key}
POST /api/v1/patients/{subject_key}/predict
GET  /api/v1/patients/{subject_key}/explanation
```

```
GET  /api/v1/audit/summary
GET  /health
```

9. Five Workstreams — Exact Responsibilities and Interfaces

Workstream 1 — NACC Data, Cohort & Clinical Pipeline

Owner: Member 1

Work:

1. Submit/track Freeze 75 data request and inspect UDS dictionary.
2. Implement MCI anchor, 24-month progression labels and exclusion rules.
3. Match candidate clinical visits to MRI dates using the shared rule.
4. Construct 2–3 visit sequences and retain real time intervals.
5. Clean the six fixed clinical variables and missingness masks.
6. Create patient-level 70/15/15 split and dataset audit.

Outputs: cohort_manifest.parquet, visits_clinical.parquet, splits.json, data_audit.json + figures

Definition of Done: Every retained patient has one label, one anchor, 2–3 ordered visits, clinical vectors and a split assignment.

Workstream 2 — MRI Feature Pipeline

Owner: Member 2

Work:

7. Parse NACC Mixed Protocol MRI analysis data.
8. Identify anatomical quantitative variables and remove non-model metadata.
9. Join imaging rows to Member 1 visit matches.
10. Apply ≤20% training missingness rule, zero-variance removal, imputation and scaling.
11. Generate feature-level missingness masks and anatomical feature dictionary.

Outputs: mri_feature_dictionary.json, mri_preprocessor.pkl, visits_imaging.parquet, mri_audit.json

Definition of Done: Every usable visit has identical-size imaging_vector[M] and imaging_mask[M], with human-readable feature names.

Workstream 3 — Multimodal Longitudinal Model

Owner: Member 3

Work:

12. Implement clinical encoder, imaging encoder, concatenation fusion, GRU and prediction head.
13. Implement variable-length sequence handling and sequence masks.
14. Implement time-step prediction loss, training loop, early stopping and checkpointing.
15. Expose stable `predict(patient_sequence)` inference interface.

Outputs: model.pt, model_config.yaml, training_history.json

Definition of Done: A real NACC PatientSequence runs end-to-end to final probability and risk trajectory without manual intervention.

Workstream 4 — Evaluation & Explainability

Owner: Member 4

Work:

16. Implement metrics, validation threshold selection and calibration.
17. Train final-visit logistic reference baseline.
18. Implement Integrated Gradients and leave-one-visit-out ablation.
19. Aggregate MRI-region, clinical-variable and modality attribution.
20. Generate final evaluation plots.

Outputs: metrics.json, baseline_metrics.json, ROC/PR/calibration/confusion figures, explanations/

Definition of Done: `explain(patient)` returns visit importance, top MRI regions, top clinical variables and modality attribution.

Workstream 5 — Platform, Backend, Frontend & Integration

Owner: Member 5

Work:

21. Create repo, Docker/dev configuration and shared contracts on Day 1.
22. Implement FastAPI backend, auth, SQLite user DB and service boundaries.
23. Implement Streamlit pages against mock API first.
24. Integrate real model/explanation services when contracts are ready.
25. Own end-to-end tests, demo mode and release packaging.

Outputs: authenticated web app, real inference integration, real explanation integration, synthetic demo mode

Definition of Done: Login → patient selection → prediction → trajectory → explanation works end-to-end through the backend API.

10. Mock-First Integration Strategy

All five members start on Day 1. No workstream waits for NACC access. The shared contract is frozen before implementation begins.

```
MockPatientSequence
  subject_key
  label
  imaging_values      [3, 20]
  imaging_mask        [3, 20]
  clinical_values     [3, 6]
  clinical_mask       [3, 6]
  delta_time          [3, 1]
  sequence_mask       [3]
```

Mock	Start	Replacement	Target date
mock_uds.csv	19 Sep	Freeze 75 UDS	23–25 Sep
mock_mri.csv	19 Sep	Freeze 75 MRI analysis	23–25 Sep

20 fake MRI features	19 Sep	Retained real MRI features M	26 Sep
Synthetic PatientSequence	19 Sep	Real model-ready sequences	26 Sep
MockModelService	19 Sep	Frozen real model service	7 Oct
mock_explanation.json	19 Sep	Real IG + ablation outputs	9 Oct
Synthetic frontend patients	19 Sep	Pseudonymized real patient views	10 Oct

11. Monorepo Structure

```
explainable-longitudinal-medical-ai/
├── README.md
├── pyproject.toml
├── docker-compose.yml
├── Makefile
├── .env.example
├── configs/
│   ├── data.yaml
│   ├── model.yaml
│   ├── training.yaml
│   └── app.yaml
├── data/
│   ├── README.md
│   ├── restricted/          # never committed
│   ├── processed/           # never committed
│   └── demo/
├── ml/nacc_progression/
│   ├── contracts.py
│   ├── data/
│   ├── model/
│   ├── training/
│   ├── evaluation/
│   └── explainability/
├── backend/app/
│   ├── main.py
│   ├── api/routes/
│   ├── auth/
│   ├── db/
│   ├── schemas/
│   └── services/
├── frontend/
│   ├── Home.py
│   ├── api_client.py
│   ├── pages/
│   └── components/
├── artifacts/
│   ├── checkpoints/         # restricted/private
│   ├── preprocessors/       # restricted/private
│   └── demo/
├── reports/
│   ├── dataset_audit/
│   ├── metrics/
│   ├── figures/
│   └── final_report/
├── scripts/
├── tests/
│   ├── unit/
│   └── integration/
├── docs/
└── .github/workflows/ci.yml
```

12. Git Workflow and Ownership

- Protected branches: `main` and `develop`. No direct commits.
- Feature branches: `feature/data-cohort`, `feature/mri-pipeline`, `feature/model`, `feature/evaluation-xai`, `feature/platform`.
- Each member owns their primary folders, but shared contracts are reviewed jointly.
- Pull request → automated tests → one team review → merge to `develop`.
- `develop` must remain runnable daily. Stable release candidates are promoted to `main`.
- Final tag: `v1.0.0` on 15 October 2026.

12.1 Shared contract rule

The file `ml/nacc_progression/contracts.py` defines `PatientSequence`, `PredictionResult`, `ExplanationResult` and `PatientSummary`. No member changes these interfaces alone. Any contract change requires agreement from Data, Model and Platform owners.

13. Data Security and Deployment Rules

- Never commit NACC CSVs, processed participant-level tables, NACCIDs, ID mappings, patient exports, `.env`, user DB files or restricted model artifacts.
- `.gitignore` must cover `data/restricted/**`, `data/processed/**`, `artifacts/checkpoints/**`, `.env`, and `*.db`.
- Private research mode uses authorized NACC data.
- Public/demo mode uses synthetic patients and the same backend/frontend interfaces.
- The frontend never displays original NACCIDs.

14. Gantt Roadmap — 19 September to 15 October 2026

Dates are indicative but the release date is fixed. Work starts on mocks immediately; real NACC data replace mocks without changing interfaces.

Owner	Workstream	19–21 Sep	22–25 Sep	26–30 Sep	1–5 Oct	6–10 Oct	11–15 Oct
M1	Data/Cohort	Mock cohort +	Real UDS + labels	Final cohort +	Leakage/QC fixes	Support integration	Final audit
M2	MRI features	Mock parser	Real MRI schema	Final M features +	Distribution/QC	XAI labels support	Final audit
M3	Model	Model on mocks	Real-data dry run	Train/tune fixed	Freeze final model	Integration support	Repro check
M4	Eval/XAI	Metrics mocks	Eval scaffolding	Validation +	IG + ablation	Final test metrics	Report figures
M5	Platform	Repo + auth + mock UI	API integration	Real PatientSequence	Model service	Real XAI + UI	QA + release

14.1 Milestones and gates

Date	Milestone	Required evidence	Gate
21 Sep	Mock system operational	Login → mock patient → mock prediction → mock explanation; neural network runs on synthetic	M0
25 Sep	NACC access / schema checkpoint	Real UDS and MRI analysis available or clarification submitted; first real joined rows inspected.	M1
30 Sep	Real model-ready dataset	Cohort, MRI features, clinical vectors, patient-level splits and first	M2
5 Oct	Training complete	Fixed architecture trained; validation results stable; no architecture	M3
7 Oct	Model freeze	`final_model.pt`, config, preprocessing artifacts and selected	M4
10 Oct	Full research path integrated	Real prediction + risk trajectory + explanations reachable through	M5
13 Oct	Release candidate	End-to-end tests, security checks, docs and figures complete.	RC1
15 Oct	Final release	Tag `v1.0.0`; private research mode + synthetic demo mode delivered.	FINAL

14.2 Dependency order

Day 1 contracts + mocks
↓
NACC UDS + MRI analysis ingestion
↓
Cohort + visit matching + feature preprocessing
↓
Real PatientSequence
↓
Fixed multimodal GRU training
↓
Frozen model + threshold
↓
Integrated Gradients + visit ablation
↓
FastAPI real services
↓
Streamlit real integration
↓
End-to-end QA → v1.0.0

15. Final Acceptance Criteria

Area	Acceptance condition
Data	Reproducible cohort rules, patient-level split, no leakage, audit report, real NACC data never committed.
Model	Fixed architecture trains and infers on real PatientSequence data; final checkpoint reproducible from config.
Performance	AUROC and required secondary metrics reported on untouched test set; reference baseline included.
Explainability	MRI-region, clinical-variable, visit and modality attribution available for a selected patient.
Backend	Authenticated FastAPI endpoints work for patient retrieval, prediction, explanation and audit summary.
Frontend	Authenticated Streamlit interface displays patient history, prediction, risk trajectory and explanations.
Demo	Synthetic demo mode runs without restricted NACC data.
Engineering	Unit/integration tests pass; `develop` and `main` are stable; README and setup instructions are complete.

16. Current Official Sources

- **NACC Freeze 75 — September 2026:** https://www.naccdata.org/publish-with-nacc-data/doi-digital-object-identifier/freeze-75-september-2026/
- **NACC Data Front Door / Quick Access:** https://www.naccdata.org/data-front-door
- **NACC Data Request Process:** https://www.naccdata.org/data-request-process/
- **NACC Data File FAQ:** https://www.naccdata.org/data-file-faq
- **NACC Longitudinal Neurocognitive and Clinical Phenotype Data:** https://www.naccdata.org/about-nacc-data/longitudinal-neurocognitive-and-clinical-phenotype-data
- **NACC Imaging Data:** https://www.naccdata.org/about-nacc-data/imaging-data
- **NACC Researcher's Guide:** https://www.naccdata.org/the-nacc-researchers-guide